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A BLOCKCHAIN CONSENSUS PROTOCOL BASED ON MULTIDIMENSIONAL PARAMETER CREDIT

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ABSTRACT

With the rapid development of the industrial Internet, it has greatly improved industrial production efficiency. However, challenges such as information asymmetry and the inadequacy of a single indicator in supporting supply-demand matching have emerged. To address the challenges, we propose a credit-based consensus protocol for the Industrial Internet. It introduces a credit model that combines multidimensional to achieve supply-demand matching and incorporates regulatory preferences for flexible global regulation. Furthermore, a credit-based Proof-of-Work difficulty adjustment method is proposed to enable distributed industrial Internet collaboration. The effectiveness of the proposed approach is demonstrated through experiments, and future research directions are discussed.

Key Words— Industrial Internet; Blockchain; Credit Model; Consensus Protocol

1. INTRODUCTION

As the Industrial Internet evolves rapidly, it has continuously surpassed its role as a platform and medium for value transmission, gradually transforming into an important tool to improve industrial production efficiency. However, on the other hand, the requirements for industrial collaboration are also increasing. The industrial Internet's value chain is complex, involving a large number of different enterprises, integrating various infrastructures such as equipment, networks, and platforms, which makes collaboration challenging. Traditional industrial collaboration guided by a few indicators such as price cannot facilitate the rapid exchange and comparison of other information, such as production status and delivery conditions. In business competition, participating enterprises that excessively pursue the prominence of a single product feature will inevitably compromise on other features, leading to typical competition dilemmas. This eventually leads to slim margins, compromised quality, and may not be ideal for the long-term growth of the industry.[1].

Conventional centralized platforms face issues of trust and expense. For instance, most current industrial Internet platforms are usually set up and controlled by a single original equipment manufacturer (OEM), primarily aiming to consolidate enterprises within their own supply chains. The main problem faced by such platforms is that the cost-effectiveness of intelligent transformation for enterprises along the entire value chain towards a single platform is not high. On the other hand, in collaborative scenarios, how to reasonably match demand and production capacity based on multidimensional indicators is also a challenge.

To address the aforementioned challenges and difficulties, we propose a credit-based consensus mechanism for Industrial Internet Blockchain. By introducing multidimensional parameters, establishing a credit system, and dynamically adjusting the difficulty of the Blockchain Proof-of-Work (PoW) algorithm, the proposed mechanism enables distributed collaboration and multidimensional matching of demand and supply. The participating enterprises in industrial Internet collaboration serve as nodes in the value Blockchain, incorporating regulatory requirements to establish a fair and self-reinforcing collaborative environment based on open protocols. The specific contributions of this paper are as follows:

- A Credit Model incorporates multidimensional indicators and regulation preference, allowing for more accurate and comprehensive matching of demand and supply in industrial Internet collaboration.
- A Credit-Base Proof-of-Work Consensus Protocol that can dynamically adjust the difficulty of the algorithm based on a credit model, ensuring efficient and fair participation of nodes in the Blockchain network.

2. RELATED WORK

The rapid development of industrial Internet has greatly contributed to the prosperity of the social economy. However, due to factors such as information asymmetry, the issue of multi-party credit has become increasingly prominent, which hinders the healthy development of industrial Internet ecosystems. The study[2] points out the significant advantages of

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using blockchain technology to establish an e-commerce credit system. It presents a framework for a blockchain-based e-commerce credit system and analyzes application scenarios in the field of blockchain e-commerce, including traceability, payments, and supply chain management (SCM). Hyperconnected[3] builds a blockchain technology system that supports trusted connections to protect data sovereignty, aiming to facilitate multi-party collaboration in the industrial Internet. In the study[4], a consensus mechanism based on node behavior credit is proposed by combining blockchain and the Internet of Things (IoT). It utilizes a directed acyclic graph (DAG) to categorize node behavior in the blockchain system as either good or bad, and calculates the node's credit score based on its behavior to influence the efficiency of Proof of Work (PoW).

Our previous work[1] proposed a cross-chain industrial Internet collaboration framework, consisting of a data collaboration chain and a value transaction chain. The goal of the value transaction chain is to use blockchain technology to ensure the security of value circulation transactions while introducing new evaluation parameters and mechanisms beyond price, enabling transactions and market regulation.

3. OVERVIEW

In order to incorporate multidimensional indicators into industrial Internet collaboration, we propose a credit-based Blockchain consensus mechanism for the Industrial Internet. As shown in Figure 1, in the proposed Blockchain network, both the demand side and the supply side are participants in the network.

The demand side sends transactions representing specific product or service demands into the Blockchain network. Nodes on the supply side, acting as miners in the Blockchain, package the transactions and generate blocks. Transactions packaged by specific nodes indicate that the corresponding demand is satisfied by that node, achieving demand matching.

The supply-side packaging nodes use a credit-based dynamic-adjustment Proof-of-Work (PoW) consensus protocol, where the difficulty of block generating is dynamically controlled by a credit model, influenced by factors such as the degree of demand-supply matching and preferences for global regulation preference. Through the consensus process, this Blockchain system enables industrial Internet collaboration based on multidimensional indicators, facilitating effective information flow, and catering to the diverse requirements of both production collaboration and e-commerce transactions.

On the other hand, by introducing global regulatory control preferences, rigid and uniform management measures can be reduced, enabling dynamic, organic, and efficient industrial collaboration regulation, while lowering regulatory costs.

4. CREDIT MODEL

As mentioned in Section 3, we propose a credit model to introduce multidimensional indicators into the collaboration process.

Based on the blockchain framework showed in Figure 1, the multidimensional indicators can be defined as below.

$$\vec{P} = (p_1, p_2 ... p_n) \tag{1}$$

where n represents the number of dimensions for the multidimensional indicators. p_i represents the i-th dimensional indicator, which can be product metrics, quality, project duration, logistics, etc., and vector $\vec{P}$ is used to collectively describe the demands preference or the service capabilities of the demand and the supply sided in the blockchain framework across multiple dimensions.

A demand that can be for certain products or service can be described as below.

$$D = (\vec{d}, price, gas) \tag{2}$$

where $\vec{d} = (d_1, d_2, ... d_n)$ represents the demand preference of D for the indicators, d_i represents the preference for the indicator p_i. $price$ denotes the price or economic return corresponding to the demand, while gas represents the resources consumed to fulfill that demand. gas is used to limit the number of demands met by an individual supplier, ensuring the system's distribution and fairness.

For a supply-side node S, its service capability regarding multidimensional indicators $\vec{P}$ can be represented as follows:

$$S = (\vec{s}) \tag{3}$$

where $\vec{s} = (s_1, s_2, ... s_n)$ represents the demand indicator fulfillment capability of the supply node, and s_i represents the fulfillment capability of the indicator p_i.

To introduce global regulatory control into the system, the preference can be described as below.

$$\vec{R} = (r_1, r_2 ... r_n) \tag{4}$$

where r_i represents the global regulatory preference parameter for p_i. For a specific demand D, the combination of the global regulatory preference can yield a new demand preference $\vec{d'}$ from the original $\vec{d}$, which can be represented as:

= (d$_1 r_1, d_2 r_2 ... d_n r_n$)(5)

Based on the above equations, the credit of the supply-side node S for the demand D with the global regulatory preference can be represented as the product of demand preferences and service capability.

$$C(S, D, R) = \vec{d'} \cdot \vec{s} = \sum_{i=1}^{n} d_i r_i s_i \tag{6}$$

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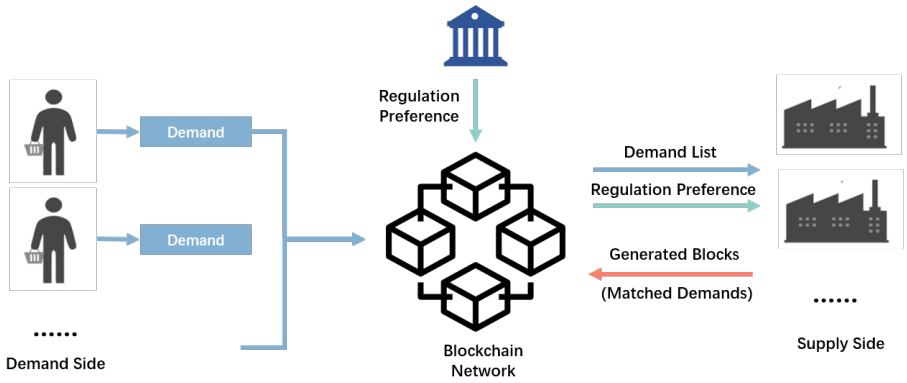

Fig. 1. Overview

5. CREDIT-BASE PROOF-OF-WORK CONSENSUS PROTOCOL

Building on the credit model outlined in Section 4, a PoW consensus framework is developed, drawing inspiration from the Bitcoin system. As shown in the Figure 2, the system consists of two types of nodes:

- **Demand nodes**: representing participants from the demand side, they publish demands in the form of blockchain transactions, verify the validity of blocks, but do not possess mining capabilities, similar to lightweight nodes in BitCoin.
- **Supply nodes**: representing participants from the supply side, they compete through PoW to package the demands published by demand nodes into blocks, achieving demand-supply matching, similar to mining nodes.

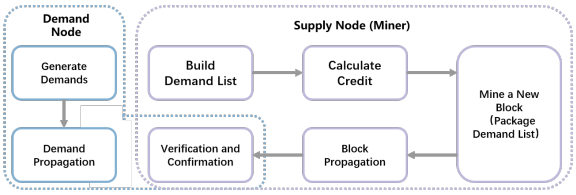

Fig. 2. Consensus Protocol

5.1. Protocol for Demand Nodes

Demand nodes are responsible for proposing demands and broadcasting them in the blockchain network. They are not involved in block generation, so the protocol is relatively simple and mainly includes the following steps:

1. **Generate Demands**: The demand node generates demand D, as definded in Equation 2, representing its specific product or service requirements.
2. **Demand Propagation**: The demand node sends D as a blockchain transaction into the blockchain network for other supply nodes to match and package into blocks.
3. **Verification and Confirmation**: Validate the newly generated blocks in the network and collaborate with the supply side after confirming the block that contains their own demand.

5.2. Protocol for Supply Nodes

Supply nodes are similar to mining nodes in the Bitcoin network. Based on their own service capabilities, supply nodes package demands as blockchain transactions and compete for the right to generate blocks using the PoW algorithm. The hash difficulty is dynamically adjusted according to the credit model to achieve demand-supply matching. Once a demand is successfully packaged into a new block by a supply node and confirmed by the network, it indicates that the order for that demand is obtained by the supplier. The specific protocol mainly includes the following steps:

1. **Build Demand List**: Supply node S selects a suitable list of demands $\vec{D}$ to be packaged into a new block based on its own service capabilities, satisfying the following conditions:

$$\begin{aligned} &\vec{D} = (D_1, D_2, ...D_k) \\ &D_i = (\vec{d_i}, price_i, gas_i) \\ &\sum_{i=1}^{k} gas_i < G \end{aligned} \tag{7}$$

Where G is the gas limit for demands contained in a single block. The total gas value of demands that a supply node can package in a block must not exceed G, which is a global variable used to ensure system distribution and fairness.

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2. **Calculate Credit**: According to Equation 6 from Section 4, S can calculate the credit indicator in combination with $\vec{D}$, which can be represented as:

$$C(S,\vec{D}) = \frac{\sum_{i=1}^{k} C(S,D_i)}{k} = \frac{\sum_{i=1}^{k}\sum_{j=1}^{n} d_{ij} r_j s_j}{k} \tag{8}$$

$$Diff(S,\vec{D}) = \frac{\alpha}{C(S,\vec{D})} \tag{9}$$

Where α is the scaling coefficient of mining difficulty, which can also be considered as the benchmark value for mining difficulty. The tuning of mining difficulty can be achieved through an approach akin to modifying the target value in Bitcoin's PoW algorithm. The variable α can be fine-tuned according to the block creation time within the network to control the network's overall mining difficulty. Node S can optimize the construction of $\vec{D}$ to improve the value of the credit indicator C and increase the success rate of mining, while satisfying the gas constraints as shown in Equation 7. On the other hand, the regulatory preference $\vec{R}$ can amplify the importance of a specific dimension parameter in the multidimensional indicator $\vec{P}$, such as environmental evaluation criteria, to encourage participating enterprises to enhance their capabilities in that dimension and achieve overall regulation objectives.

3. **Mining a New Block**: Supply node S packages the transactions corresponding to $\vec{D}$ into a new block using the PoW algorithm.

4. **Block Propagation**: The new block is sent into the blockchain network for other nodes to verify.

5. **Verification and Confirmation**: Similar to the demand nodes, the supply nodes verifie the new block and add it to their own longest chains.

6. EVALUATION

To showcase the efficacy of the suggested credit-based consensus model in balancing supply and demand, we employ BlockSim[5], a blockchain simulation platform. Based on the Bitcoin model, we modified the computational difficulty component of the block consensus and implemented the Credit-based Proof-of-Work Consensus Protocol mentioned in Section 5. The important parameter settings for the experiment are summarized in the following Table 1.

The experimental blockchain network consists of 5 supply nodes, and we introduced a 5-dimensional multidimensional indicator which can vary based on the application scenario.

Table 1. Evaluation Setup

Parameter	Setup
Indicator Dimension	5
Node Number	5
Average block propagation delay	0.42 sec
Simulation Time	20000 sec
Transaction Number	100
G: Gas limit for a block	3.0
$Epoch$: Number of simulation runs	3
gas: for a transaction	$X \sim Exp(1.0)$
$price$: for a transaction	$X \sim N(gas, (0.3 \cdot gas)^2)$

The service capabilities of each node can be represented by a modified normal distribution, as follows:

$$\vec{s} \sim N(1.0, 0.2^2) \tag{10}$$

To reflect the varying capabilities of different nodes in serving different dimensional indicators, each node has an additional performance of 0.7 in one dimension, as $para0$, $para1$, $para2$, $para3$, and $para4$ showed in Figure 3.

From the demand perspective, the parameter requirements for each dimension also follow a normal distribution and can be represented as:

$$\vec{d} \sim N(1.0, 0.2^2) \tag{11}$$

To reflect the demand for specific indicators, each demand also has an additional parameter bonus of 0.7 in specific dimensions.

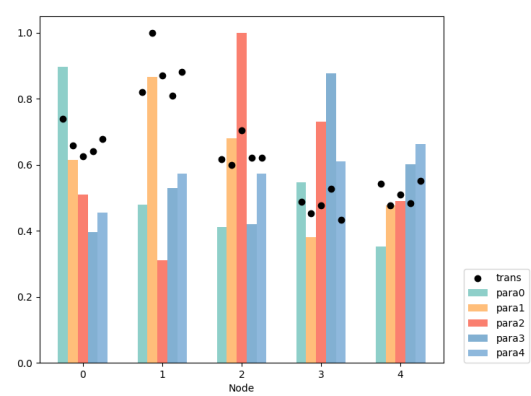

Fig. 3. Simulation Result without regulatory preference

As shown in Figure 3, the y-axis in the graph represents the normalized results of each group of data. $trans$ represents the normalized sum of all demand transactions packaged by each node across different dimensions. From the graph, we can see that each node has an advantage in the dimensions

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where they have additional parameter bonuses. This demonstrates that the consensus mechanism is capable of achieving demand-supply matching across multiple parameter dimensions in a distributed manner.

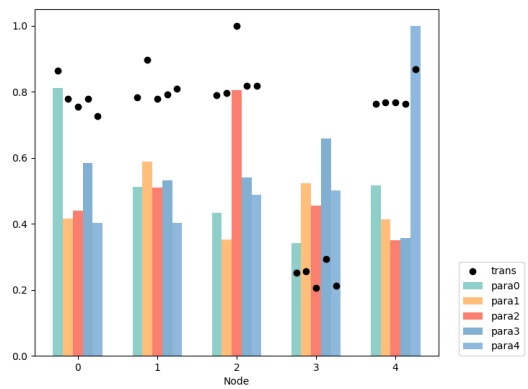

Fig. 4. Simulation Result with regulation preference

In Figure 3, a uniform regulatory preference of $\vec{R} = (1, 1, 1, 1, 1)$ is employed. To verify the impact of regulatory preference, Figure 4 adopts a regulatory preference setting of $\vec{R} = (1, 1, 2, 1, 1)$. The results show that Node 3, which has an additional preference bonus, achieves the highest satisfaction in each dimension for the fulfilled demands and completes the largest number of 62 out of 256 demands. This demonstrates the effectiveness of regulatory preference in collaborative coordination.

7. FUTURE WORK

We proposes a credit-based consensus mechanism that introduces multi-dimensional indicators into the blockchain network, addressing key issues in industrial Internet collaboration. Building upon this research, future exploration can be conducted in the following research directions:

- **Research Credit-based Proof-of-Stack (PoS) Protocol**: The credit-based consensus Protocol is particularly suitable for PoS (Proof of Stake) scenarios. Further research can focus on dynamically adjusting the stake of nodes and incorporating collaborative behavior into the consensus mechanism, aiming to achieve more flexible regulation.

- **Incentive mechanisms**: Improve incentive mechanisms that encourage participants to contribute their resources and collaborate actively within the cooperation system.

By further exploring these research directions, we can advance the understanding and application of the credit-based consensus mechanism, paving the way for more efficient and effective industrial Internet collaboration.

8. CONCLUSION

We present a credit-based consensus protocol for industrial Internet , addressing challenges through multidimensional parameters and dynamic difficulty adjustments. It enables distributed collaboration and multidimensional matching of demand and supply. Participating enterprises act as nodes in the value blockchain, incorporating regulatory requirements for a fair collaborative environment. The proposed credit-based consensus protocol was experimentally validated to effectively match demand and supply across multiple dimensions in a distributed manner. The contributions include a Credit Model for accurate matching of demand and supply, and a Credit-based Proof-of-Work Consensus Protocol ensuring efficient and fair node participation in the blockchain network.

9. ACKNOWLEDGEMENTS

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